

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Definition: Whether the expected output modality matches regional deployment needs and accessibility.

Justification

Output form is text-classification labels evaluated by macro-F1 with results averaged over three runs and standard deviations reported [Q93, Q96]. This is a standard, well-documented evaluation methodology and matches the deployment's text-to-label format at surface level. However, the deployment requires confidence scores for routing thresholds and multi-label outputs for messages spanning multiple departments [elicitation Q4]. LTZGLUE reports only macro-F1 with no calibration metrics (e.g., ECE, Brier score), no per-class precision/recall thresholds, and no multi-label evaluation methodology. Invalid LLM outputs are simply discarded [Q90, Q91], which masks the operational reality that malformed outputs in deployment must be handled (escalated). OF is MODERATE priority. The form is text-to-text-label as required, but the metric set and label cardinality do not match the deployment's operational form.

Strengths

- Text-output modality matches deployment [Q93]
- Multi-run evaluation with standard deviations reported for encoder models [Q89, Q96] supports meaningful comparison
- Hyperparameter search via Bayesian optimization with early stopping is methodologically sound [Q136, Q138]
- Class-balanced loss for imbalanced tasks [Q92] is a conscientious design choice

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Text output matches deployment, but the deployment additionally requires confidence scores and multi-label outputs [elicitation Q4], which the benchmark's macro-F1 single-label evaluation does not assess.

OF-2: Not applicable — deployment is text-only [regional context infrastructure_notes.deployment_modality], and benchmark is text-only [Q93].

OF-3: Civil servants are professionally literate [regional context target_population.civil_servant_profile]; literacy is not a constraint. Citizens have 99.0% internet penetration [WEB-7]. No accessibility-driven output modality concerns.

ID	Evidence
WEB-7	99.0% internet penetration — confirms text-output channel is universally accessible

OF-4: External-validity gap: macro-F1 alone cannot validate the calibration and multi-label behaviors required for routing escalation logic; invalid outputs being discarded [Q90, Q91] hides a deployment-relevant failure mode.

Information Gaps

- No calibration metrics (ECE, reliability diagrams) reported

- No multi-label evaluation methodology defined
- No analysis of malformed-output rate as an operational signal

Requires Expert Verification

- Specification of operational confidence thresholds and escalation-rate targets by deployment owners
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Remediation

Gap 1: Macro-F1 only; no calibration metrics; invalid outputs are silently discarded; no multi-label evaluation

Recommendation: Add calibration metrics (ECE, reliability diagrams), multi-label metrics (micro/macro-F1, hamming loss, subset accuracy), and treat malformed-output rate as an explicit deployment KPI rather than discarding such cases.